Comb: Project Guide

BongoHive internal finance paper trail. How to run it, what it does, and how to sign in.

1. What Comb does

Comb replaces three Excel workbooks (Expense Submission, Procurement form, Staff Advance & Retirement) with one web app. People request, approve, attach evidence to, and seal every transaction **before** it reaches the books. Tagline: *Every kwacha gets a cell. A cell is filled with approvals and evidence, then sealed.*

Tool	Role	Comb's relationship
Xero	Accounting, bank feeds, tax	Comb exports paid bills as a CSV for Xero. It never keeps the books.
ClickUp	Tasks and projects	Not replaced. Quotes live on the record they support.
Bank / Airtel / MTN	Moving money	Comb never sends money. Finance pays, then records the reference.
Payroll	Salaries	Out of scope.

2. The three workflows

• **Expense:** draft, manager, finance, final approval, payment, evidence check, sealed.

• **Staff advance:** draft, manager, finance, final approval, payment, payee confirms, retirement, finance check, approval, settle balance, sealed.

• **Procurement:** draft, manager, finance, quotes, check, authorise, purchase order, goods received, three-way match, final approval, payment, evidence check, sealed.

3. Built-in rules and controls

• Quote thresholds: up to K10,000 one quote; to K20,000 two; above K20,000 three; above K100,000 open tender.

• Expenses above K10,000 must go through procurement. Overdue advances block new ones (can be switched off).

• Nobody approves their own request and nobody signs twice in one approval gate. Three different people are needed: reviewer, finance, final approver.

• Sign-offs record person and time and never change. Sealed records are locked in the database, with a fingerprint to detect outside tampering.

• A voucher cannot pay more than the value received (three-way match: PO, goods received, invoice).

• Review flags: split purchases, reused files, duplicate invoices, budget overruns, weak justifications, overdue advances.

• Every document prints or saves as PDF; **Full audit pack** bundles a whole transaction.

4. Where AI is used

Optional. Claude reads receipts, invoices and quotes to prefill forms, and drafts value-for-money justifications. It never approves, pays or moves a request. Without an `ANTHROPIC_API_KEY` everything

works and people type details by hand.

5. How to run it

Requires Node 24 or newer (this machine has v24.13.1). From the project folder:

```
npm install
npm run demo   # sample data, http://localhost:4300
npm start     # real data and real passwords
npm test      # 54 automated tests (all pass)
```

6. Sign-in details

Demo mode (`npm run demo`) has **no passwords**. The sign-in page shows one-click buttons for each sample role:

Person	Role	What to test
Seiko Mwenya	finance	Finance review, payment, vouchers, Admin, Xero export
Lukonga Lindunda	director	Final approvals
Simunza Muyangana	director	Second director approvals
Maumo Mubila	staff	Create requests, advances, procurement
Mapalo Lukashi	staff	Second requester

Real mode (`npm start`): on the first run the terminal prints a one-time password for the finance lead (username `seiko.mwenya`), shown once, and you must change it on first sign-in (minimum 10 characters). Usernames are `firstname.lastname`. Add people and set their passwords in Admin. If locked out: `npm run reset-password seiko.mwenya`.

7. Before real use

• Host behind HTTPS on a server you control (set HOST and TRUST_PROXY).

• Back up the data/ folder daily. It holds the database and evidence, unencrypted.

• Test the Xero CSV with a small batch against your own Xero template.

• Confirm exchange rates, thresholds and electronic-approval acceptance with finance, auditors and donors.

• Run alongside the spreadsheets on one project for a month first.

8. Project layout

Path	Purpose
src/rules.js	Money and policy rules
src/workflow.js	Stages, who may act, separation of duties, sealing
src/flags.js	Review flags
src/ai.js	Claude document reading and drafting
src/app.js, src/views/	Web routes, pages and printable documents
seed/, test/, brand/	Reference data, tests, logos